

Mechanisation and Safety

Science

May, 2026

Mechanisation and Safety (A36582)

Short Title: Mechanisation and Safety
Faculty Engineering
Department Science
Cluster Engineering
Author TWOODCOCK
Credits 5

Level: Introductory

Description of Module / Aims

The module will enable the student to acquire knowledge and awareness to work safely. It will enable students to understand the principles of risk assessment in the workplace. The student will acquire the principles, knowledge and skills needed for pesticide and fertiliser application. The module will enable students to understand the basic principles of engines, hydraulics and electrics and carry out basic maintenance on tractors. The student will understand the safe operation of tractors and machinery.

Programmes

MECH-0005	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	2 / 3 / M
MECH-0005	BSc in Agriculture (WD_SAGUL_D)	1 / 1 / M
MECH-0005	BSc in Forestry (WD_SFORS_D)	1 / 1 / M

Indicative Content

- Health & safety
- Risk assessment and manual handling
- Power Units
- Hydraulics and pneumatics
- Electrics
- Pesticide & fertiliser application
- Safe tractor operation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe safe working methods and practices in accordance with associated legislation.
2. Complete manual handling, a risk assessment and Health and Safety statement for a workplace situation.
3. Explain working principles and maintenance of power units in agriculture / forestry.
4. Describe working principles of hydraulics / pneumatics.
5. Explain basic electrics for farm use.
6. Describe the operation/ calibration of a pesticide and fertiliser applicator.
7. Demonstrate an understanding of safe operation of tractors and machinery.

Learning and Teaching Methods

- Lectures.
- Self-directed learning
- Practical demonstrations.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,3,4,5
Continuous Assessment	50%	
<i>Practical</i>	50%	2,3,5,6,7

Assessment Criteria

- <40%:** Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.
- 40%-49%:** Will be able to show a basic knowledge of the key learning outcomes including animal structures, functions and living processes as well as be able to perform simple laboratory experiments with animals.
- 50%-59%:** As well as the above the student will be able to demonstrate an understanding of some of the complexities of the topics and predict nutrition requirements of the animal and potential nutritional disorders. The student will also be able to perform good quality experiments with animal tissues.
- 60%-69%:** In addition, the student will demonstrate more detailed knowledge of the all topics including and demonstrate an ability to evaluate detailed knowledge acquired in the module to new material. The student will also be able to perform very good quality experiments with animal tissues and microbiology related to animals.
- 70%-100%:** The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, predict nutrition requirements of the animal and potential nutritional disorders and show the ability to evaluate and relate topics that overlap all of the learning outcomes. The student will also be able to perform high quality experiments in microbiology and with animal tissues.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Bell , P. _Safe Use of Pesticide Products_. Dublin: Teagasc, 2020.
- Ryan , M. _Tractor Maintenance_. Dublin: Teagasc, 2019.
- Teagasc, T. _Farm Safety and Farm Assurance_. 2022: Teagasc, 2022.
- Teagasc, T. _Mechanical and Manual Pesticide Application_. Dublin: Teagasc, 2019.

Supplementary Material

- "Health and Safety Authority Agriculture Documents." https://www.hsa.ie/eng/publications_and_forms/publications/agriculture_and_forestry/
- "RSA Important standards for agricultural vehicles such as tractors and category R trailers." <https://www.rsa.ie/road-safety/road-users/agricultural>

Requested Resources

- WORKSHOP: Kildalton Agricultural College Equipment and Machinery